

# Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

☎ (587) 879-4626 | ✉ [tkuipers123@gmail.com](mailto:tkuipers123@gmail.com) | 🌐 [tkuipers.ca](http://tkuipers.ca) | 📷 [tkuipers](#) | 🌐 [tyler-kuipers](#)

## About Me

I've spent 10 years building backend systems, with the last 3 focused on architecture and technical leadership at AMA. What pulls me toward your team is the mission. I spent a year at Arterys building backend services for a medical-imaging AI platform, and getting back to healthcare, especially the kind aimed at people the system underserves, is what I want next. Day to day I own architecture decisions, write the RFCs and design docs behind them, run deep code reviews, and mentor a 12-person team. On my own I built a self-service developer platform that took environment provisioning from two to three weeks down to two hours, the sort of dev-experience and tooling work your team cares about. TypeScript is home territory, and I work well with a lot of autonomy and little ceremony.

## Work Experience

### Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Own and maintain 4 critical REST APIs serving 5 internal teams and external partners. Designed API contracts, approve all new endpoint additions, and maintain API design standards documented in Confluence. APIs process \$4.3M in annual transactions. (Python, REST, OpenAPI, TypeScript)
- Built an event-driven pipeline that imports an external partner's policy data into our platform: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite PolicyCenter API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)
- Architected technical infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, \$4.3M annual revenue, 2.8M insured trips processed, 99.9% uptime since launch. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, Terraform, TypeScript)
- Built self-service developer platform solo that reduced environment provisioning from 2-3 weeks to 2 hours (60x improvement). Adopted by 15 personnel across 3 teams, created 120+ environments. Unblocked major initiatives by enabling rapid iteration and production debugging via snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Architected modernization of legacy document generation system into containerized microservices on EKS. 4-5 services (with sidecars) on a 3-5 node cluster, deployed via Helm. Datadog observability for failure count, success count, generation time, and storage time. Leading 2-3 engineers on the project. (Kubernetes, EKS, Docker, Helm, Datadog, C#)
- Set technical standards for 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Member of Architecture Steering Committee. Help newer developers navigate legacy codebases and trained developers on TDD methodology.

### Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

### Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Java, Gosu)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

### Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

### Lockheed Martin Canada

ASSOCIATE SOFTWARE ENGINEER

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

|                                                                                                                                                                      |                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <div><div>The Open University</div><div>MSC. COMPUTING (SOFTWARE ENGINEERING)</div><div>Currently pursuing Master's in Computing at The Open University.</div></div> | <div><div>Milton Keynes, England (Remote)</div><div>May 2024 - Ongoing</div></div> |
| <div><div>The King's University</div><div>B.S. IN COMPUTER SCIENCE</div></div>                                                                                       | <div><div>Edmonton, Alberta</div><div>Sep 2012 - Dec 2016</div></div>              |

Skills

|                                                                   |                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Backend Architecture &amp; APIs</div></div>             | <div>Design Python and TypeScript services on AWS that run in production. Own and maintain 4 critical REST APIs with OpenAPI design standards, serving 5 internal teams and external partners and processing \$4.3M in annual transactions. Architected an event-driven serverless system that has held 99.9% uptime since launch and processed 2.8 million events.</div> |
| <div><div>Healthcare Backend</div></div>                          | <div>Built and maintained backend services that ran medical AI algorithms and returned results to the platform at Arterys, a medical-imaging company. Improved algorithm processing turnaround using Redis caching and AWS infrastructure tuning. Monitored production via Elasticsearch and Kibana, catching and resolving issues before customer impact.</div>          |
| <div><div>Cloud &amp; Infrastructure</div></div>                  | <div>Production AWS across multiple accounts (Lambda, EKS, SQS, RDS, IAM) with Terraform for infrastructure as code. Built and maintain a developer platform (60x faster provisioning), and a production EKS cluster running cert-manager, Datadog, Karpenter, and Linkerd.</div>                                                                                         |
| <div><div>Technical Leadership &amp; Domain Ownership</div></div> | <div>Set technical standards for 12-person team through code reviews, RFC authorship, PR checklists, and API design guides. Member of AMA's Architecture Steering Committee. Drive cross-team architectural decisions including error handling and sanitization standards. Own technical domains from architecture through production.</div>                              |

Technologies

|                                                  |                                                                                           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------|
| <div><div>Backend Languages</div></div>          | <div>TypeScript, Python, Java, Node.js, Bash, C#</div>                                    |
| <div><div>Architecture &amp; APIs</div></div>    | <div>REST APIs, OpenAPI, Microservices, Event-Driven Architecture, Async Workflows</div>  |
| <div><div>Databases</div></div>                  | <div>PostgreSQL, Aurora/RDS, Redis</div>                                                  |
| <div><div>Cloud &amp; Infrastructure</div></div> | <div>AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Terraform, Kubernetes, Docker, Helm</div> |
| <div><div>Observability</div></div>              | <div>Datadog, Elasticsearch, Kibana, Metrics Design, Alerting</div>                       |